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Mechanical origin of shape memory performance for crosslinked epoxy networks

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ABSTRACT

The mechanical principle of microstructural change governing the shape formation and restoration process of shape memory epoxy (SME) was analyzed on a subcontinuum scale. A series of processes to program and operate the microstructure of highly crosslinked networks across the phase transition temperature range was implemented in molecular dynamics simulations. This elucidated the mechanisms by which the chemical composition of resins and crosslinkers, the degree of orientation of each chain, and the topology of the entire network characterize the shape memory behavior of the system. The strategy for analyzing the mechanical behavior of molecules by classifying them into translation, rotation, and deformation based on the classical kinematic framework was particularly effective in clarifying the structure–property relationship. The results showed that, during the shape programming process, each molecular component of the SME was rearranged to different levels depending on its stiffness, forming local residual stresses. The principle leading to shape recovery as the subsequent thermal load removes residual stress and the resulting change in the mechanical anisotropy of the entire system were also successfully comprehended.

1. Introduction

Shape memory polymers are promising organic materials with the ability to memorize large deformabilities depending on the environment set by the user [1,2,3]. When an external force is applied with sufficient heat, the polymer enters a temporary dormant state, and its microstructure is programmed. The memorized microstructure returns to its original state when reheating occurs, leading to macroscopic mechanical deformation [4,5]. Among the shape memory polymers reported thus far, shape memory epoxy (SME) with a highly cross-linked network is the simplest and most widely studied [6,7]. A network structure composed of chemical covalent bonds has various advantages, including high chemical resistance, excellent mechanical properties, satisfactory thermal durability, and outstanding shape fixation and restoration performance [8,9,10,11]. In particular, by adjusting the crosslinking density and network topology, the temperature at which the shape memory effect is expressed and the operating characteristics, such as its deformability and recovery rate, can be finely controlled [12,13,14,15,16,17]. Therefore, SMEs have high applicability in various fields, such as the operating parts of soft robots [18,19,20], space-deployable structures [21,22,23], and stimulus-responsive inks for 4D printing [24,25,26,27,28,29].

The core of the SME mechanism is to induce microstructural programming and macroscopic mechanical deformation by going back and forth between the rubber phase and glass phase based on the glass transition temperature (T_g) [30,31,32]. The entire process experienced by SME materials during their life cycle can be divided into two major phases: 1) the programming process that determines the macroscopic shape of the material and 2) the process of restoring the microstructure to its initial state by thermal stimulation. At high temperatures above T_g, the mobility of the polymer chain is assured, enabling macroscopically flexible molding. When it returns to the glass phase through subsequent cooling, it is fixed in a temporary shape by the strong bonding force of the epoxy network. Such temporary shapes begin to break as heat is reapplied above T_g, and the associated macroscopic shapes also return to their pre-programmed amorphous state.

Therefore, the functionality of SMEs is determined by the competition between the thermally induced mobility of polymer chains and structural rigidity of the networks that make up the microstructure. In

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this regard, several attempts have been made to quantitatively identify the factors that determine the structure–property relationship at the molecular level. Diani et al. [33] and Amini et al. [17,34] intensively studied the morphological properties of polymer chains during the shape memory effect. By parameterizing curling and kinking as a function of the radius of gyration, they interpreted the correlation between conformational changes and the macroscopic shrinkage behavior of polymer chains geometrically. Studies conducted by Ghobadi et al. [35] and Wick et al. [36] addressed the structural changes in molecular components during thermomechanical cycles. These studies established that atomic energy accumulated in certain molecular components is a key factor in triggering the shape memory effect. The distribution of internal energy during deformation was interpreted using changes in the potential energy components formed by the bonding of several atoms, such as bond stretching, angle, and dihedral angle. However, the mechanical behavior of SMEs with highly cross-linked networks is dominated by structural properties that a combination of several atomic bonding properties cannot explain. At the molecular scale, SMEs have a mixture of regions that mainly form collective translational motion and regions that exhibit high deformability, depending on the type and bonding relationship of the molecules constituting the crosslinked network. In particular, the formation of molecular orientation in a specific direction and material anisotropy are accompanied by the processes of shape memory and recovery of the SME. Nevertheless, a theoretical approach that clearly explains the correlation between material anisotropy, macroscopic mechanical strain, and cross-linked network structural properties at the molecular scale has not been developed.

Recently, we developed a methodology to examine how the topology of the cross-linked epoxy and the associated subcontinuum parameters determine its mechanical properties [37]. The mechanics of the materials can be identified from the analysis of the virial stress of each atom formed by the change in the potential energy of the microstructure under a mechanical load. In other words, when a complex polymer structure changes in state due to external stimuli, the flow, accumulation, and loss of mechanical stress at the molecular scale can be quantitatively understood through a classical mechanical approach. In the case of epoxies, by classifying the strain energy contribution according to molecular bonding and structural properties, the topology of the network formed by resins and crosslinkers can be correlated with their load transfer characteristics.

Based on the recently developed subcontinuum analysis methodology, this study identifies the molecular scale factors governing the shape memory behavior of epoxy for the first time. The thermodynamic environment experienced by the SMEs throughout their lifetime was implemented from an all-atom model, and the mechanism of strain energy distribution, storage, and release in molecular networks during each process was systematically evaluated. The approach makes it possible to comprehensively investigate the origin of the shape memory effect, shape recovery rate, and effective mechanical properties expressed by the microstructure during molding and operation. In particular, we propose that the response characteristics of the SME networks during the shape memory cycle can be hierarchically understood from the three parameters: translational motion, rotational motion, and internal deformation of the molecular components. The mechanical principles by which the distribution of strain energy applied to molecular components, the change in network shape, and the formation of residual stresses determine the mechanical anisotropy of materials are thereby clarified.

The present study provides a theoretical understanding of the influence of the chemical composition of molecules and the conformational properties of the resulting microstructure on the macroscopic shape memory behavior. By demonstrating the molecular scale factor that governs the driving force for inducing shape memory, which is difficult to discover experimentally, we provide a guide to precisely design the operation process of shape memory epoxy according to the user's intention.

2. Description of the shape memory characteristics using MD simulations

To investigate the microstructural changes during the shape memory process, a shape memory epoxy model was prepared using all-atom molecular dynamics (MD) simulations. Fig. 1**(a)** shows the atomistic configurations of the epoxy components used in this study. The diglycidyl ether of bisphenol A (DGEBA) and 4,4-diaminodiphenylmethane (DDM) was selected as the resin and hardener for the microstructure, respectively. A total of 250 DGEBA and 125 DDM molecules were randomly dispersed in a cubic cell box, where periodic boundary conditions were applied in all directions. To prevent undesired interference between the inserted molecules, the cell was initially set to have a low density of 0.2 g/cc. Then, the epoxy microstructure was gradually densified by reducing the length of all axes of the cell box equally. For every 0.2 g/cc increase in density, geometry optimization was performed on all constituent atoms using the conjugate gradient method with a cut-off value of 0.001 kcal/mol. After the system's density reached 1.0 g/cc, the microstructure was relaxed using an NVT ensemble at 300 K for 1 ns, followed by an NPT ensemble at 300 K and 1 atm for 1 ns. All timesteps for time integration were set to 0.5 fs, except for the parts separately specified.

The steps for forming crosslinked epoxy networks are shown in Fig. 1 **(b)**. First, all the atoms participating in the crosslinking reaction (the terminal carbon atom of the epoxide ring and the nitrogen atom of the amine) were identified. The closest distance between the neighboring reactive atoms was monitored, and a covalent bond was created if the distance was less than the set reaction radius. The reaction radius was initially set to 4.0 Å and then increased in 0.5 Å steps until the crosslinking ratio of the entire microstructure reached 60%. Whenever a covalent bond was generated, the aromatic ring of the epoxide group of DGEBA was opened while removing the redundant hydrogen attached to the DDM. Also, each time the crosslinking reaction was completed at a given reaction radius, a short equilibration of the NVT ensemble at 300 K for 0.3 ps was performed to release the undesired potential energy resulting from the newly created covalent bonds. A single macro network was created in the unit cell because the crosslinking reaction was performed across periodic boundaries. Finally, the epoxy microstructure was further relaxed using the NVT ensemble at 300 K for 1 ns and NPT ensemble at 300 K and 1 atm for 2 ns. The complete model of the crosslinked shape memory epoxy is shown in Fig. 1**(c)**. Representative physical properties calculated from prepared cross-linked epoxy microstructure models are listed in **Table S1**, along with relevant well-known experimental values. The results confirmed that the structural, thermal, and mechanical properties of the model before entering the shape memory programming stages were consistent with the experimental observations.

The shape memory effect of the model was characterized using MD simulations of two processes: programming and recovery. The detailed steps of each process and their conditions are shown in Fig. 1**(d)**. The first process involves applying a mechanical load to the epoxy in a rubbery state to form a temporary shape, which is further subdivided into the following five steps: (i) A crosslinked epoxy model fully equilibrated at 300 K and 1 atm was prepared. (ii) The model was heated to $T_g + 30$ K using an NPT ensemble of 1 atm for 2 ns to achieve high deformability. When the epoxy network was exposed to high temperatures, the temperature of the shape memory epoxy model was adjusted using the Nosé–Hoover thermostat with a coupling time constant of 10 fs. (iii) The unit cell box was stretched uniaxially in the z-direction until the maximum strain reached 30%, referring to programming experimental conditions of shape-memory epoxy of the same composition [38]. With the time step used for time integration set to 0.5 fs, a total MD running time of 6 ns is provided, resulting in a strain rate of 5×10^7 /s.

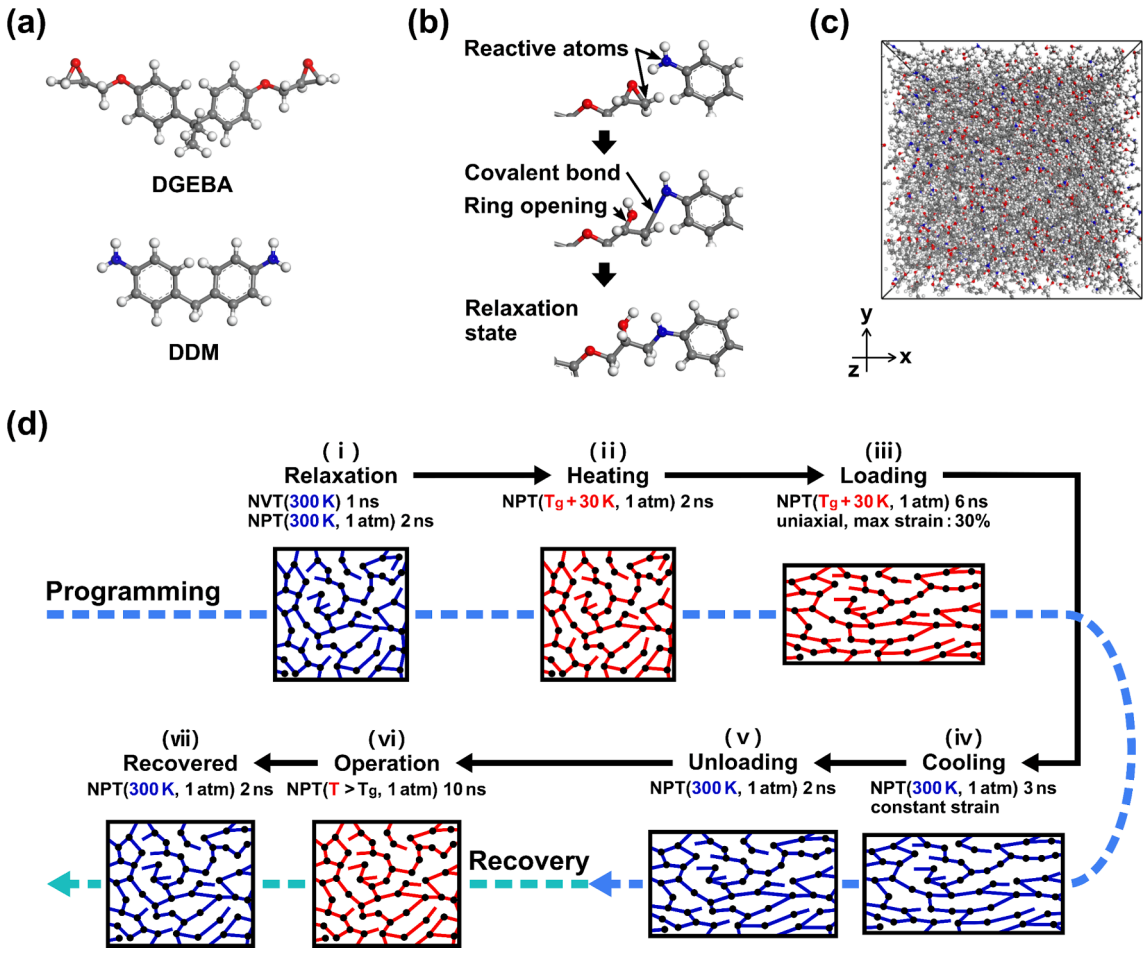


Fig. 1. An MD simulation process for the evaluation of shape memory properties of crosslinked epoxy networks. (a) Molecular structures of the epoxy resin (DGEBA) and hardener (DDM). (b) Description of crosslinking reaction of epoxide group in DGEBA with the amine group in DDM. (c) The representative periodic unit cell of the crosslinked epoxy microstructure. (d) The process of fixing the epoxy network into an operable shape (from steps (i) to (v)) and triggering the shape memory effect (from steps (vi) to (vii)).

(iv) The model was gradually cooled down to 300 K at a rate of 0.12 K/ps while maintaining the applied strain. Atmospheric pressure was applied only in the direction perpendicular to the strain using a Nosé-Hoover barostat. (v) The deformation constraint of the unit cell box in the z direction was removed. The model was further relaxed using an NPT ensemble at 300 K and 1 atm for 2 ns. On the other hand, in the second process, strain recovery occurs by the reheating of the programmed epoxy network, and this process is subdivided into two steps. (vi) Under the NPT ensemble, the model has been tested for its behavior at various operating temperatures ranging from T_g + 30 K to T_g + 120 K. The sequence of response times according to molecular motion properties does not change regardless of the reheating temperature conditions (**Fig. S1**). Also, all the temperatures considered do not approach the thermal decomposition temperature of epoxy materials (647 K to 657 K) [39,40]. Therefore, T_g + 120 K was chosen as the smallest of the temperatures at which recovery rates>0.9 could be achieved within a time integration of 10 ns. (vii) After the operation was completed, the epoxy model was cooled down to 300 K for comparison with the initial model before the shape memory cycle. During a series of programming and recovery processes, the microstructural changes and evolution of the internal stress were thoroughly discussed.

Note that the processes of model equilibration, shape formation, heating and cooling applied to all-atom MD simulations are all completed under time integration within nanoseconds. Although the absolute time period is incomparably shorter than the rate for evaluating the shape memory effect in the experimental environment, the arrival of the thermodynamic equilibrium state at each step was clearly confirmed. In particular, the change in the apparent structure of the modeled system and the associated physical properties closely follow those of the well-known SME. In other words, the amount of time integration computation considered is long enough that the energy change applied to the model at each step can be redistributed to the entire epoxy network microstructure and updated to a new thermodynamic equilibrium. In particular, during the forcible deformation of the system by shape memory programming and reheating, there was no local energy accumulation, dissipation, or distortion inside the microstructure. The validity of the thermodynamic state obtained for the model is discussed separately later in the relevant section.

The other simulation details are as follows. Atomistic modeling of the epoxy microstructure was performed using the commercial MD simulation package Materials Studio 2020 (Dassault Systèmes, France). Initial construction of the epoxy microstructure was performed using the Amorphous Cell module and subsequent densification and stabilization of the SME were performed using the Forcite Plus module. The distance-based epoxy crosslinking simulation was automated using a perl script. The atomic positions and bonding relationships of the crosslinked model were transferred to the open-source code LAMMPS (Sandia Lab., USA). Subsequently, shape memory cycle application and mechanical tests were performed on the model. All interatomic interactions of the molecules were described using a polymer consistent force field (PCFF) [41].

3. Results and discussion

3.1. Prediction of the glass transition temperature

The glass transition temperature of the model was estimated to accurately simulate the epoxy shape-memory effect. Changes in mean square displacement (MSD) and density were obtained during cooling-down simulations, following the methods previously proposed by the authors [42]. An NPT ensemble at 500 K and 1 atm was applied for 3 ns to the epoxy model equilibrated at room temperature. Maintaining the isobaric condition, the model was gradually cooled down by 10 K every 0.5 ns until the temperature reached 300 K. The MSD curves were obtained during the last 50 ps of each temperature stage using the following equation:

$$MSD = \frac{1}{N}\sum_{i=1}^{N}\left[\vec{r}_i(t) - \vec{r}_i(0)\right]^2 \quad (1)$$

where $\vec{r}_i(t)$ indicates the position vector of i -th atom at time t and N is the total number of atoms. The measurement was repeated five times, and the results were averaged. As shown in Fig. 2**(a)**, a clear gap in the MSD curve between 380 K and 390 K can be observed. The intermolecular non-covalent bonding strength and mobility formed by the epoxy networks change rapidly in this temperature range, indicating the transition from the rubbery to the glassy state of the system. Based on the MSD results, the density change profile was divided into regions of 300 K to 380 K and 390 K to 500 K, as depicted in Fig. 2 **(b)**. The data for each range were fitted using linear regression ($R^2 > 0.97$), and the glass transition temperature value was obtained from the point of intersection of the two lines. The estimated value of the glass transition temperature was 387.7 K, which is in excellent agreement with the reported experimental (386.7 K) and theoretical (392 K) values [38,43].

3.2. Structural characterization during the shape memory process

The microstructural changes in the shape memory process of the shape memory epoxy are described with the kinematics of each molecule. The external stimuli (i.e., the mechanical and thermal loadings) applied for shape programming irreversibly alter the distribution of internal energy in the molecular components [44]. The material properties of the programmed bulk structure can be interpreted as the result of changes in the internal energy distribution of the molecular microstructures. Taking the above into consideration, we evaluated the structural effects experienced by epoxy microstructures in the process of imparting shape memory properties as three independent mechanical behaviors: translation, rotation, and deformation. First, the translational motion of each component can be evaluated by tracing the molecule's center of mass (T).

$$T = \frac{1}{M}\sum_{i}^{n} m_i d_i \quad (2)$$

where m is the mass of the molecule, and d is the distance from the origin of the unit cell. M is the total mass of the system, and n is the total number of molecules. Second, because the rotational motion is related to the relative angular change of the main alignment direction of each molecule and a specific axis, it corresponds to the molecule's orientational order parameter (S) as follows.

$$S = \left\langle \frac{3cos^2\theta - 1}{2} \right\rangle \quad (3)$$

where θ indicates the angle with the programmed axis, and the bracket symbol, $\langle\ \rangle$, indicates the average of the molecules included. The details of calculating the angle formed by the molecular components are described in **Fig. S2** of the Supplementary Data. Finally, the mechanical deformation of the epoxy network causes internal stresses by changing the distribution of interaction forces between molecules. Therefore, the deformability of each molecular component can be represented in terms of the atomic stress (σ) based on the virial theorem as follows [45].

$$\boldsymbol{\sigma} = \frac{1}{V}\left(-\sum_{i}^{N} m_i\left(\boldsymbol{v}_i \times \boldsymbol{v}_i^T\right) + \frac{1}{2}\sum_{i}^{N}\sum_{j\neq i}^{N} \boldsymbol{r}_{ij} \times \boldsymbol{F}_{ij} \right) \quad (4)$$

where m_i and $\boldsymbol{v}_i$ are the mass and velocity of the i -th atom, and $\boldsymbol{r}_{ij}$ and $\boldsymbol{F}_{ij}$ are the relative position and the interatomic force between the i -th and j -th atoms. N and V indicate the total number of atoms and total volume of the unit cell, respectively.

Fig. 3**(a)** shows the correlation between the temperature and density of the epoxy system during the shape programming. Under a given thermostat, the temperature fluctuations show only negligible deviations within 10 K, confirming that the SME microstructure has an instantaneous internal thermal equilibrium with temperature change (see **Fig. S3**). The density profile is inversely proportional to the applied temperature because of the thermal expansion properties of the thermoset polymers. At each step of the temperature change, the system density change converges sufficiently; therefore, the modeled system can be considered to be in a quasi-static state during the entire shape programming process. As the epoxy transformed into a shape with shape memory characteristics, the center of mass, orientational order, and atomic stress response of the molecular components were monitored as shown in Fig. 3**(b)**. During the heating step to ensure high deformability of the material, the center of mass and orientational order showed no appreciable change. However, when an external load was applied simultaneously at a high temperature, both the center of mass and orientational order profiles increased linearly in the forming direction. This shows that the deformation energy caused by mechanical loading is rapidly propagated to the entire microstructure at a given strain rate and

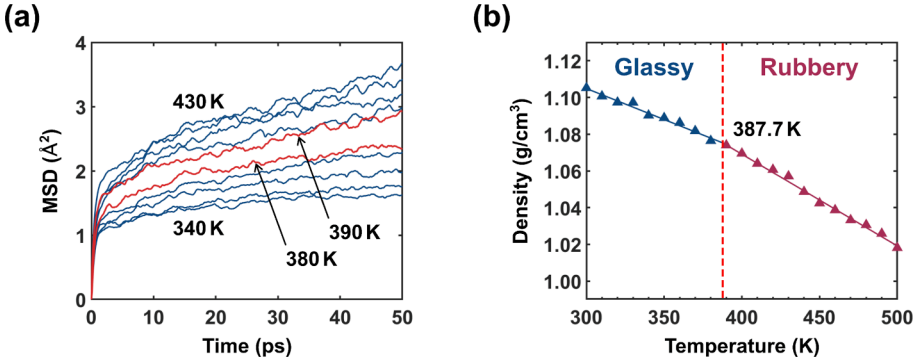


Fig. 2. Estimation of the glass transition temperature of the modeled shape memory epoxy. (a) The MSD curves in the temperature range of 340–430 K. (b) The density change profile with temperature. Glassy and rubbery regions were divided based on the MSD results, and the corresponding glass transition temperature value was also specified.

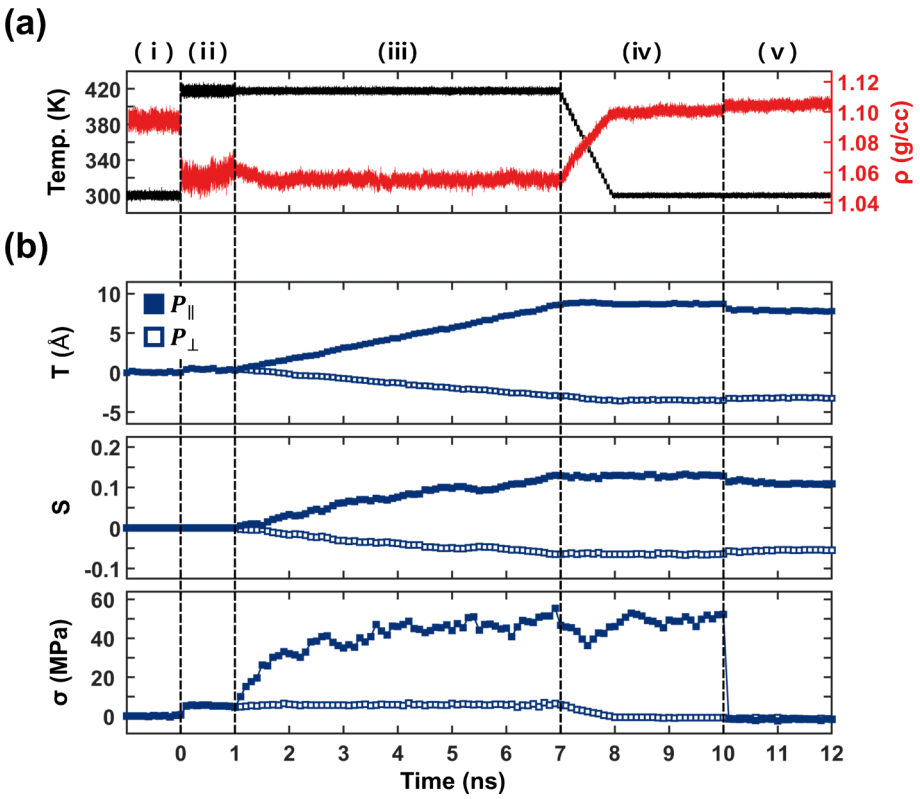


Fig. 3. Characterization of microstructural change of shape memory epoxy during the shape programming process. (a) Applied temperature for shape-memory programming and the associated change in the density of the entire system. (b) Changes in the mass center, orientational order, and stress under thermodynamic ensembles.

the target thermodynamic environment is maintained. The components in the direction perpendicular to the load were relatively decreased, and thus, the anisotropy of the material increased. Such a significant rearrangement of network components was maintained during the subsequent cooling and unloading (with a slight spring back), resulting in a permanent modification of the epoxy network. The retention rate of the molded structure can be quantified by the fixity ratio (R_f) and is defined as:

$$R_f = \frac{\varepsilon_f}{\varepsilon_m} \tag{5}$$

where ε_m is the maximum tensile strain applied to the unit cell, and ε_f is the effective strain remaining after the constraint along the tensile direction is removed. After shape programming was completed, the fixity ratio of the SME model reached 0.89 (see **Fig. S4** of Supplementary Data), which agreed with the experimentally observed value [46,47].

To further investigate the mechanical role of each molecular component in the system, the effective atomic stress was analyzed for each programming step. As shown in Fig. 4**(a)**, the crosslinked epoxy networks were classified into DDM and DGEBA, and DGEBA was further subdivided into aromatic and epoxide groups. The virial stress applied to each group was calculated using Eq. (4), and calculations were performed considering only the atoms included in the group. Other details of the stress analysis for each group in the polymer microstructure were performed following the method proposed by Kim and Choi [37].

Fig. 4**(b)** shows the atomic stresses distributed in each epoxy component. When the model was heated to a high temperature, the atomic stresses of the system increased slightly because the kinetic energy of the entire component increased. During the subsequent loading process, the stress value increased monotonically over the entire (elastoplastic) range along the forming direction. The generated stresses remained large until cooling to room temperature but were quickly and completely released in the final unloading stage [48]. Therefore, little residual stress remained during shape-memory programming of the entire epoxy network.

However, it is important to note that the recovery of the system to the zero-stress state is only apparent, and this is because the DDM and DGEBA form residual stresses in opposite directions. Specifically, when an external force is applied above the glass transition temperature, most of the deformation energy accumulates in the stiff parts of the molecules. In contrast, the flexible parts (i.e., the hydrocarbon chains connecting the rigid aromatic core and epoxide groups) have good deformability, so that no atomic stress is created by stretching. The behavior was almost maintained until the system reached stage (iv) where it cooled to room temperature. However, as the mechanical load was removed in stage (v), a compressive force was generated in the system with a spring back in the forming direction, and most of the energy was absorbed by the hydrocarbon chain. The asymmetric energy storage of each part of the microstructure deepens the anisotropy of the material, providing a means of storing actuation forces that can be triggered by subsequent stimuli. The above discussion is dominated by components parallel to the direction of the mechanical load.

Fig. 5**(a)** shows the density profile of the unit cell during the shape recovery process according to the application of a high temperature. As in the previous (programming) stages, the density of the epoxy network was inversely proportional to the supplied thermal energy and showed an immediate convergence at each step (see **Fig. S5** for an enlarged view of the temperature equilibrium profile).

The microstructural changes under the thermal equilibrium of the system exposed to a high temperature of T_g + 120 K were analyzed during the recovery stages (results for different temperature conditions are presented in **Fig. S1**). Fig. 5**(b)** shows the changes in the microstructural properties of the molecules constituting the system during the recovery process. In a high-temperature environment above the glass transition, the kinetic energy of the polymer system becomes sufficient

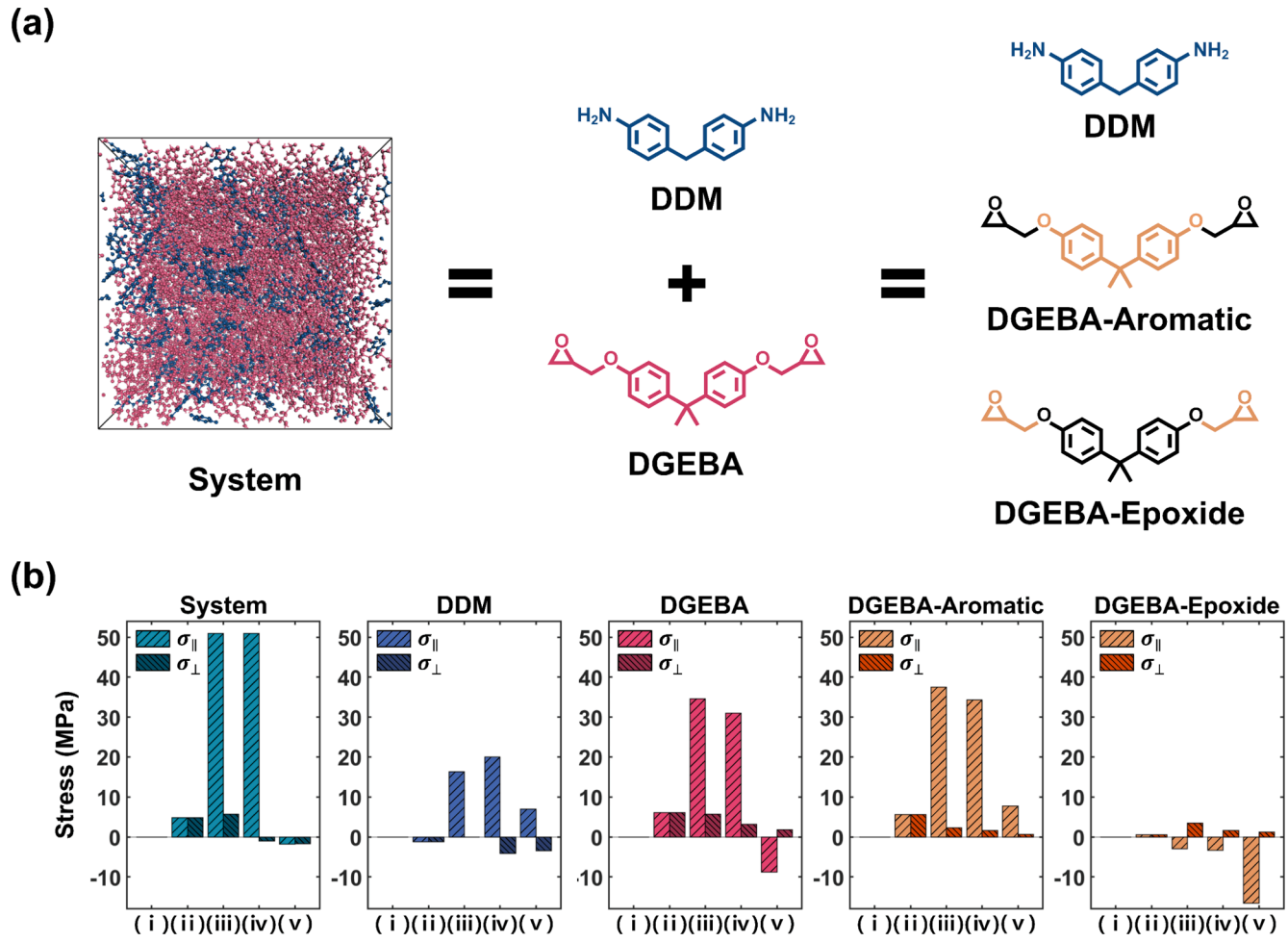


Fig. 4. Molecular level classification of the modeled microstructure. (a) The procedure for subdividing the components of the resin and hardener that constitute the crosslinked epoxy. (b) Atomic stress distributions of the classified molecular components at each stage of programming.

to break the restraints caused by non-covalent bonds between interchains, causing the mobility to increase remarkably (see Fig. 2**(a)**). The results suggest that the increase in mobility restored the length of the entire system, which had been increased during programming, to its original state. Therefore, in the programming stage, the center of mass had a clear gap between the direction of tension and the direction perpendicular to it, but the deviation between the two gradually disappeared owing to heating [49]. In the case of the orientational order, the speed at which the deviation disappears is faster, and after approximately 8.5 ns, the system becomes close to isotropic. The recovery rate to the initial shape at the operating temperature is quantified by the recovery ratio (R_r) and is defined as:

$$R_r = \frac{\varepsilon_m - \varepsilon_r}{\varepsilon_m} \tag{6}$$

where ε_r is the strain in the shape memory direction of the unit cell after the high temperature exposure is complete. After exposure to a high temperature for 10 ns, the recovery ratio for the unit cell length reached 0.93 (see **Fig. S6**), which is consistent with the experimentally reported macroscopic recovery level [50,51]. It should be noted that the quantitative matching of the experimental values of the recovery ratio, together with the previous fixity ratio, is the decisive evidence that the microstructural model covered in this study follows the shape memory characteristics very well (see **Supplementary Text 1** for relevant details). Finally, the atomic stress profile coincided with the applied temperature according to the kinetic term of the virial stress, and its response was very immediate.

Recall that a significant redistribution of stress among the molecules occurred during the shape-memory programming stages. After programming, the tensile force remained in the aromatic rings of DDM and DGEBA, with the relatively flexible side chain of DGEBA compensating for it, resulting in a stress balance for the entire system. The results shown in Fig. 6 suggest that the residual energy distributed to each component is completely released during the heating operation. By supplying a level of energy at which the thermal load can break the conformation formed by the polymer network, the energy state of the microstructure is shifts away from the local minimum, facilitating structural reconstruction [52,53]. In particular, the internal stresses of DDM, DGEBA, and related components all disappear after the shape memory effect is activated.

Another interesting finding is that the deformation, rotation, and translation responses of the molecular components are sequential rather than simultaneous. We defined the complete recovery point for each response from the intersection of the property profiles in two directions: parallel and or perpendicular to the programmed direction. The response speed of the physical properties can be compared from the time elapsed until the recovery point is reached (see Fig. 5**(b)**. **Fig. S7** also shows the independent stress profiles of the DDM hardener and DGEBA resin.). The most immediate restoration under thermal energy was found for deformation (2.6 ns), followed by full restoration for rotation (3.5 ns), and translation (>10 ns). This is in good agreement with the observation in the previous literature [54,55] that the formation of the recovery stress is immediate and rapid, in contrast to the slow shrinkage of shape memory specimens at high temperatures.

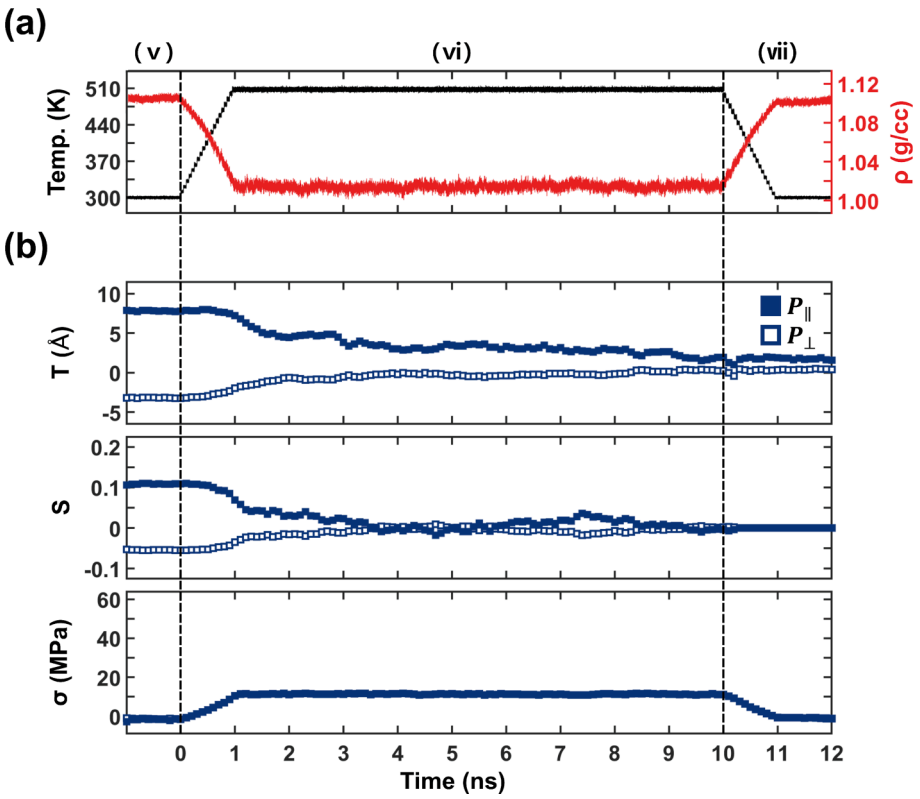


Fig. 5. Microstructural changes in response to the shape recovery process. (a) Applied temperature for shape-memory recovery and corresponding density of the entire system. (b) The response of the center of mass, orientational order, and stress to the mechanical behavior of the microstructure under thermodynamic equilibrium.

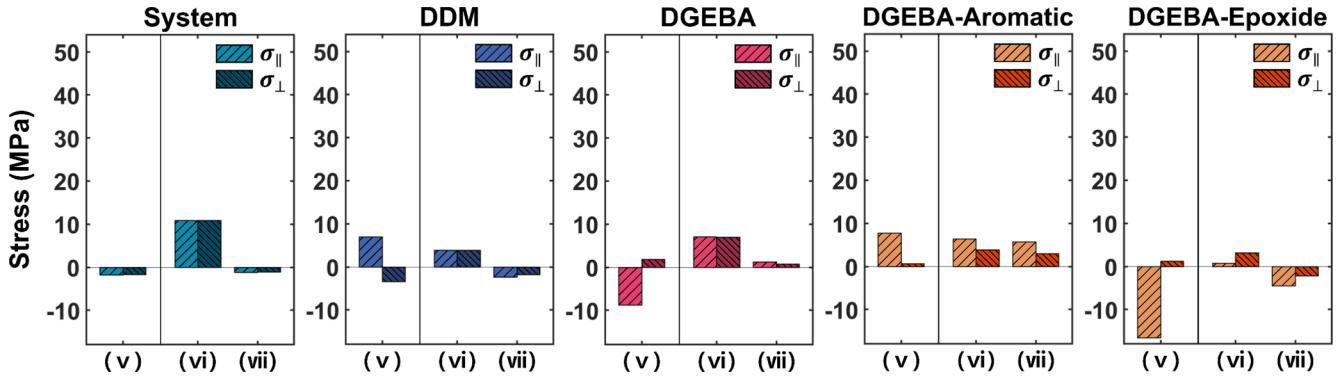


Fig. 6. Atomic stress distributions for each molecular component in the recovery stages. The stress distribution immediately after completion of the process for shape memory ability (i.e., 'stage (v)' before applying the operating temperature) is also presented for comparison.

The difference in response time is intuitively explained by the origin of the driving force that causes each mechanical mode. First, the recovery of atomic stress is due to the release of stored internal energy through the distortion of the molecular conformation [56,57]. The strength formed by covalent and non-covalent bonding of several atoms shows an extremely high sensitivity to deformation. In particular, the harmonic oscillations between covalently connected atoms exhibit very high bond strengths, so even with insignificant distortion produces considerable resilience. Therefore, when excessive thermal energy is allowed in the polymer network and the structural balance is disturbed, restoring to the initial state occurs most rapidly through conformational transformation. Second, the recovery of the rotation angle of the molecules arises from the formation of a molecular moment perpendicular to the programming direction. Because the entire network was highly crosslinked, the points of action of the thermal recovery force were mainly located at both ends of each molecule. At this time, the formation of the angular velocity of the molecule is limited by the high moment of inertia formed by the linear part of the epoxy molecule; therefore, a longer recovery time is required compared to the deformation mode. Finally, translational motion is restored when a force is applied to both ends of the molecule in the same direction. As with rotational deformation, the moment of inertia due to molecular mass and structure limits the recovery speed in this case. In addition, because the absolute displacement required to return to the initial state is much longer than that in the rotation mode, translational motion behaves much slower than the other recovery modes.

Note that the recovery process described above is distinct from contraction under compressive forces in that it is the spontaneous movement of polymer chains within the microstructure. In shape recovery behavior, chain unwinding by heat supply lowers the potential

energy of the system, but the resulting macroscopic deformation appears as a spontaneous contraction rather than a thermal expansion. This suggests that spontaneous shrinkage deformation induces significant changes in the mechanical properties of the material, which will be discussed further in the next section.

3.3. Mechanical properties

Uniaxial tensile simulations were performed to investigate the changes in the mechanical properties for each state of the shape memory epoxy. Models of a state in which only crosslinking of the resin and curing agent were made without external load (stage (i)), a state in which shape-memory properties were programmed (stage (v)), and a state in which shape recovery by reheating was completed (stage (vii)) were prepared, respectively. Uniaxial deformation was carried out at a constant strain rate of 5×10^8 /s until the maximum strain of the unit cell reached 10% under the NPT ensemble at 1 atm. The time integration for the tensile test simulation was performed with an extremely small timestep of 0.1 fs. The mechanical response of the model was calculated according to the virial theorem presented in Eq. (4), and the elastic modulus of the material was derived using the stress–strain slope at strains below 2%. In this process, two temperature conditions were considered: 300 K and a cryogenic temperature (10 K). The former was used to compare the simulation results with properties well known from the literature, and the latter was used for additional analysis of each component constituting the microstructure, which will be described later. Also, the 5×10^8 /s strain rate condition applied for uniaxial deformation is confirmed to be valid for calculating the rate-independent mechanical properties of microstructures (**Fig. S8**).

Fig. 7 shows the stress–strain curves for each major stage of shape memory epoxy. The physical properties of the amorphous model are also summarized in Table 1, along with the values reported in other studies. The mechanical properties of the amorphous model quantitatively followed the properties of typical crosslinked epoxy reported in previous experimental and theoretical reports [43,58,59,60], confirming the validity of the simulated system. Next, the formation of structural anisotropy became evident after the model was endowed with a shape-memory effect. Because the epoxy network was aligned on the molecular scale in a direction parallel to the axis of elongation, the overall mechanical properties improved in the programmed direction. Finally, after the microstructure was restored to its initial shape by heat, the structural anisotropy exhibited in the previous step completely disappeared, and it returned to a form with almost identical physical properties as those of the original amorphous state.

In order to further investigate the molecular-scale mechanics occurring in a series of processes, the deformation energy density of the subgroups constituting the epoxy network was calculated together. The stress tensors were derived separately while limiting the targets of the atoms covered by Eq. (4) to those belonging to sub-specific groups (i.e., DGEBA and DDM). The contribution of the actual workload performed by each component (u_k) in the system under a mechanical load is therefore quantitatively derived as follows:

$$u_k = \int_0^{\varepsilon_0} \sigma_{xx}^k d\varepsilon \tag{7}$$

where σ_{xx}^k is the atomic stress allocated to component k, and ε_0 is the total strain applied to the unit cell. Numerically, the value represents the linear sum of the individual atomic stresses of the atoms in the component. In other words, the deformation energy density of the entire epoxy network (u) is equal to the sum of the values assigned to each component as follows:

$$u = \sum_k u_k \tag{8}$$

The analysis method enables an intuitive comparison of load-carrying capacity of each molecular component under mechanical microstructure deformation. One challenge is that the number of atoms participating in each subgroup is extremely limited. In the process of achieving thermodynamic equilibrium of the entire system, the kinetic energy of the atoms can interfere with the analysis, making it difficult to explicitly derive properties attributable to the structural properties of the molecule. Therefore, the evaluation of deformation energy density for the subgroups was conducted only in tensile simulations in a cryogenic (10 K) environment. For the validity of the methodology and other details, see the study by Kim and Choi [37].

Fig. 8 shows the change in the mechanical response of each subgroup in the microstructure immediately after the shape memory effect was applied and after the shape recovery was completed. The strain energy density of the epoxy components was calculated using Eq. (7). The results were obtained by subtracting the value of the initial amorphous state ($u - u_{initial}$). As a result of the shape-memory programming process, it is clearly observed that all the molecular components constituting the network improved their mechanical properties along the programming direction. One interesting observation is that the magnitude of improvement in the physical properties depends on the chemical structure of the molecules. The DDM and DGEBA-Aromatic subgroups are key contributors to improving mechanical properties in the shape memory process. In contrast, the epoxide group of DGEBA had a linear structure with high deformability after crosslinking. Therefore, in the epoxide group of DGEBA, the contribution to strengthening the mechanical properties by shape-memory programming is insignificant.

Although the mechanical properties of the network in the longitudinal direction are improved by applying of the shape memory effect, the mechanical properties in the direction perpendicular to the deformation are relatively degraded. The weakening mainly occurs only in components with high flexibility, such as DGEBA-Epoxide, and molecules containing aromatic rings, such as DDM and DGEBA-Aromatic, show mechanical behavior similar to before shape programming. It is

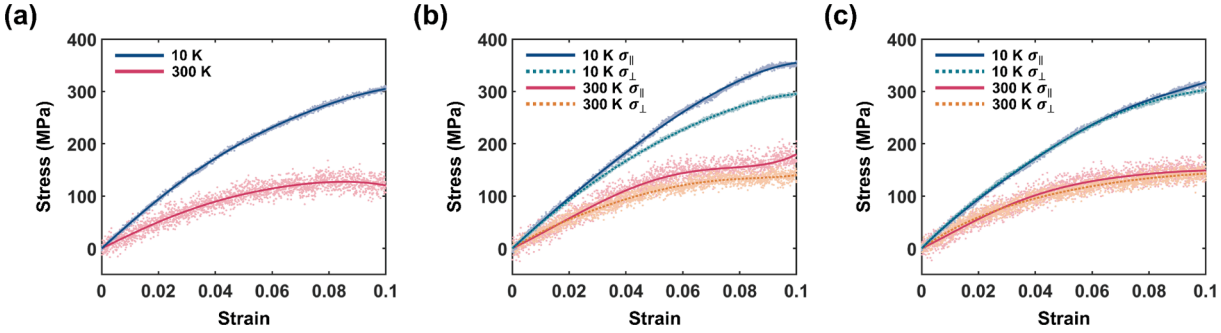


Fig. 7. Changes in mechanical properties of epoxy networks by shape memory process and shape recovery: (a) The initial amorphous state, (b) the shape-memory programming completion state, and (c) the shape recovery state by heat. In all cases, the scatter plots represent the raw data obtained from the MD simulations and the solid lines are their fifth-order polynomial regressions.

Table 1
Comparison of mechanical and structural properties with values reported from previous literature.

	Method	Composition	Crosslinking ratio (%)	Density (g/cm^3)	Young's modulus (GPa)	
					300 K	10 K
Current study	MD simulation	DGEBA	59.8	1.11	2.53	4.70
H. Nabipour et al. [58]	Experiment		N/A*	–	2.50	–
M. G. Prolongo et al. [59]	Experiment	/DDM	N/A**	–	2.50	–
W. Jian et al. [43]	MD simulation		80	1.10	3.20	–
K. Duan et al. [60]	MD simulation		60	1.17	3.17	–

* The sample was cured at a temperature above 120 °C for 8 h.
** The sample was cured at a temperature above 120 °C for 3 h.

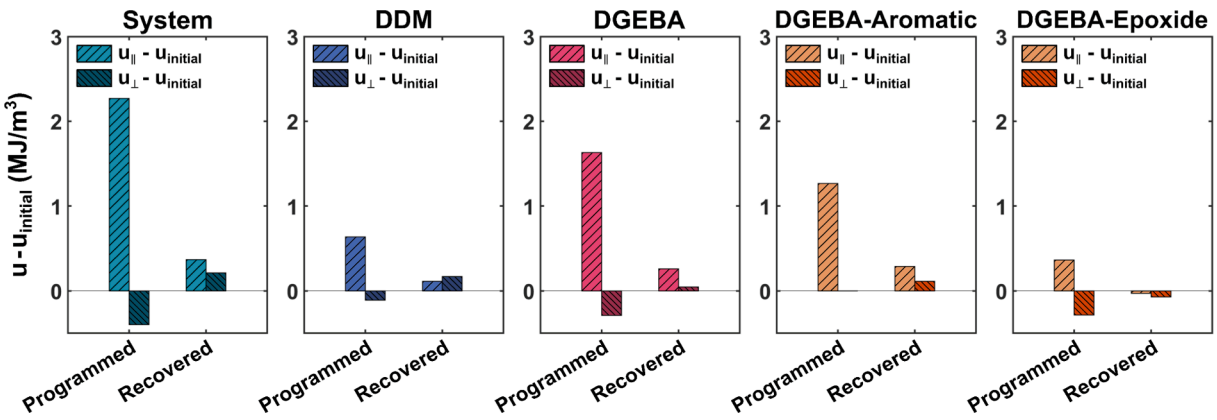


Fig. 8. Changes in deformation energy density allocated to each molecular component before and after activating the shape memory effect. All values were expressed as increments relative to the amorphous state before applying shape-memory programming. Quantitative data for each condition are presented in **Fig. S9** of Supplementary Data.

anticipated that the high π-π interaction is responsible for the ability of molecules containing aromatic rings to exhibit good load-carrying behavior. During shape-memory programming, as the structure of the epoxy network was forcibly changed, the normal vectors of the aromatic rings were oriented slightly perpendicular to the programming direction, favoring π–π interactions (see **Fig. S10** in the Supplementary Data). It is worth recalling that the energy of the π–π interaction (~4.38 kcal/mol) is much greater than the van der Waals energy between linear polymer chains (~1 kcal/mol) [61,62]. Therefore, even if these interaction pairs occur infrequently, the degradation of the mechanical properties in the vertical direction is sufficiently compensated for.

Finally, the shape recovery of the network structure by heat treatment caused the mechanical properties of the material to recover to those of the initial amorphous state. The deformation energy density against the mechanical load of the molecular components constituting the network also returned to a state similar to the initial state. Most of the residual strain energy was due to the aromatic ring of DGEBA, which is plausible because the recovery behavior of the microstructure by heat treatment did not lead to a completely amorphous state.

4. Conclusions

In this study, the changes in the structural properties of the microstructure experienced by the thermosetting polymer resin during the programming and restoration of the shape memory effect were investigated. After constructing a crosslinked epoxy network model in an all-atom MD environment, the mechanical response of the material was explored by simulating a series of processes leading to heating, tension, cooling, and reheating under thermodynamic ensembles. The response of the epoxy network at each stage was analyzed based on the molecular position change, orientational order, and virial stress distribution. The main outcomes of this study are listed below:

- The shape programming procedure permanently modulates the microstructural morphology of the epoxy network. In this process, the total stress remains in equilibrium, but a significant asymmetric residual stress remains inside the molecules constituting the epoxy network. In the direction of shape programming, the DDM has tensile residual stress and the DGEBA has compressive residual stress.
- Asymmetric residual stresses occur at the moment the shape programming is completed, and the holding load is removed. The compressive force is generated in the process of spring back and stress relief following load removal, which is mostly absorbed as the deformation energy of the hydrocarbon chain with high flexibility.
- The asymmetric residual stress serves to provide an actuation force that promotes restoration to the initial state upon reheating. The shape recovery of the network is completed by sequentially progressing three mechanical modes: release of residual stress, restoration of alignment, and translational motion of each molecule to its original position.
- The higher the complexity of the atomic structure involved in each mechanical behavior mode, the more delayed it is to contribute to the shape recovery of the overall microstructure. Such sequential behavior can be explained under a classical kinematic framework. In the order of stretching, rotation, and translation of molecules, the potential energy involved in mechanical behavior increases, while inertia decreases. Therefore, upon reheating, the stretching mode of interatomic bonds exhibits an instantaneous response, whereas the change in the center of mass of a molecule is the most insensitive.
- With reheating, the asymmetric residual stress imparted by the shape memory process is removed. The transversely isotropic elasticity, which was formed in the programming direction of the entire microstructure, is also restored to isotropic. The effective deformation energy under the mechanical load of each molecule constituting the network also returns to its properties before shape programming.

This study provides subcontinuum-scale insight into the origin of the shape memory effect of highly cross-linked epoxy networks. The mechanical effects of all major processes on the crosslinked epoxy network were systematically classified and analyzed from the imparting the shape memory effect to the completion of the shape recovery operation. The proposed model clarifies the structural factors that affect shape memory in a complex network structure and how the resulting changes in material properties occur. Therefore, this study can serve as a methodological guideline for designing the synthesis process of thermosetting resins with the desired physical properties and shape recovery performance. The effect of the presence of multifunctional nanofillers, polymer crystallinity, and blending environments incorporating heterogeneous materials on the shape memory performance of materials is also an interesting extension of this research area, which is currently underway.

CRediT authorship contribution statement

Yeongbin Kim: Methodology, Software, Visualization, Writing – original draft. **Hongdeok Kim:** Methodology, Writing – original draft. **Joonmyung Choi:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.112162.

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